

Exposé of the doctoral thesis

Exposé der Dissertation

Title of the doctoral thesis (working title):
Titel der Dissertation (Arbeitstitel):

Multimodal Airborne Detect-and-Avoid for Non-Cooperative Drones

Doctoral candidate: **Max Klemens**
Promovierender

Supervisor: **Herr Dr. Loesel**
Erstbetreuer

Secondary Supervisor: **Prof. Dr. Christian Seidel**
Zweitbetreuer

Doctoral College: **Künstliche Intelligenz/Informatik**
Promotionszentrum

Doctoral candidate
Promovierender

Supervisor
Erstbetreuer

Secondary
Supervisor
Zweitbetreuer

Contents

1	Introduction	1
2	State of the Art	3
2.1	Drone Detection and the Role of LiDAR	3
2.2	3D LiDAR Object Detection	4
2.3	Helicopter Detect-and-Avoid and Multi-Object Tracking	5
2.4	Research Gap	5
3	Research Question	6
3.1	Motivation	6
3.2	Main Research Question	7
3.3	Subquestions	7
3.3.1	Data and Observation Modeling	7
3.3.2	Tracking and State Estimation	8
3.3.3	Short-Horizon Trajectory Prediction	8
3.4	Scope and Delimitation	9
4	Methodology	10
4.1	Overview	10
4.2	Component 1: LiDAR Drone Data and Benchmark Framework	10
4.2.1	Purpose	10
4.2.2	Synthetic LiDAR Data Generation	11
4.2.3	Dataset Format and Splitting	12
4.2.4	Baseline Detection Models	12
4.3	Component 2: Tracking and State Estimation from Sparse LiDAR Detections	13
4.3.1	Purpose	13
4.3.2	Tracking-by-Detection Pipeline	13
4.3.3	Uncertainty Handling	14
4.3.4	Tracking Evaluation	14
4.4	Component 3: Short-Horizon Trajectory Prediction	15
4.4.1	Purpose	15
4.4.2	Model-Based Prediction Baselines	15
4.4.3	Learning-Based Prediction Models	15
4.4.4	Prediction Uncertainty	16
4.4.5	Trajectory Prediction Evaluation	16
4.5	Experimental Procedure	17
4.6	Expected Contribution	17
5	Research Plan	18

6	Publication Plan	19
6.1	Key Findings	19
6.2	Implications	20
6.3	Limitations	20
6.4	Future Research Directions	21
7	Abstract	22
	Bibliography	24

1. Introduction

The increasing use of small unmanned aerial vehicles (UAVs) creates new challenges for aviation safety, especially in low-altitude airspace where helicopters frequently operate. Helicopters are commonly used for rescue missions, police operations, infrastructure inspection, and medical transport. In such environments, drones can become safety-critical obstacles because they may not carry transponders, and may be difficult to detect visually under adverse lighting, cluttered backgrounds, or high pilot workload. Detect-and-avoid (DAA) systems therefore require robust onboard perception methods that can detect, track, and predict the motion of such intruders early enough to support collision avoidance decisions [fasano2016senseAvoid](#), [skowron2019senseAvoidSmallUAS](#), [riedel2025daaStandards](#).

An onboard DAA system must perceive the surrounding airspace independently of information transmitted by the intruder. Camera sensors can provide valuable appearance information, but the depth and relative position of a small object may be difficult to estimate reliably at long distances or under unfavorable illumination. Light detection and ranging (LiDAR) offers a complementary sensing principle by actively measuring the distance to surrounding surfaces and representing them as three-dimensional point clouds. This direct geometric information is particularly relevant for DAA because collision-risk assessment depends not only on recognizing an intruder, but also on determining its position and motion relative to the host aircraft. LiDAR has therefore been extensively investigated as a sensing modality for the detection and tracking of small UAVs [hammer2018potentialLidarUAV](#), [hammer2018lidarSmallUAVs](#), [dogru2022sparseLidarDrone](#).

The use of LiDAR from an airborne platform nevertheless introduces substantial challenges. Both the host aircraft and the intruding drone may move independently, so that the sensor viewpoint, background, and observation geometry change continuously. Further more at increasing distance, a small drone may generate only a few intermittent LiDAR returns, making individual detections uncertain and requiring information to be combined over time. Reliable DAA consequently requires more than object detection in isolated sensor frames. The observations must be transformed into consistent coordinate systems, associated across time, and used to estimate the intruder's relative position and velocity despite sensor noise, ego-motion, incomplete measurements, and processing delay. The quality of this estimated state is critical for the safety of helicopter - UAV encounters.

This PhD project investigates how airborne RGB and LiDAR sensing can support the detection, and tracking of small non-cooperative drones for helicopter-oriented DAA. The research is structured as three consecutive stages. First, a real-world airborne RGB-LiDAR dataset will be developed to enable reproducible encounter evaluation. Second, this benchmark will be used to identify and address limitations in the detection and tracking of UAVs to develop a precise and efficient data-driven detection model. Third, the resulting perception outputs will be incorporated into a DAA evaluation framework to assess helicopter safety and initiate avoidance maneuvers.

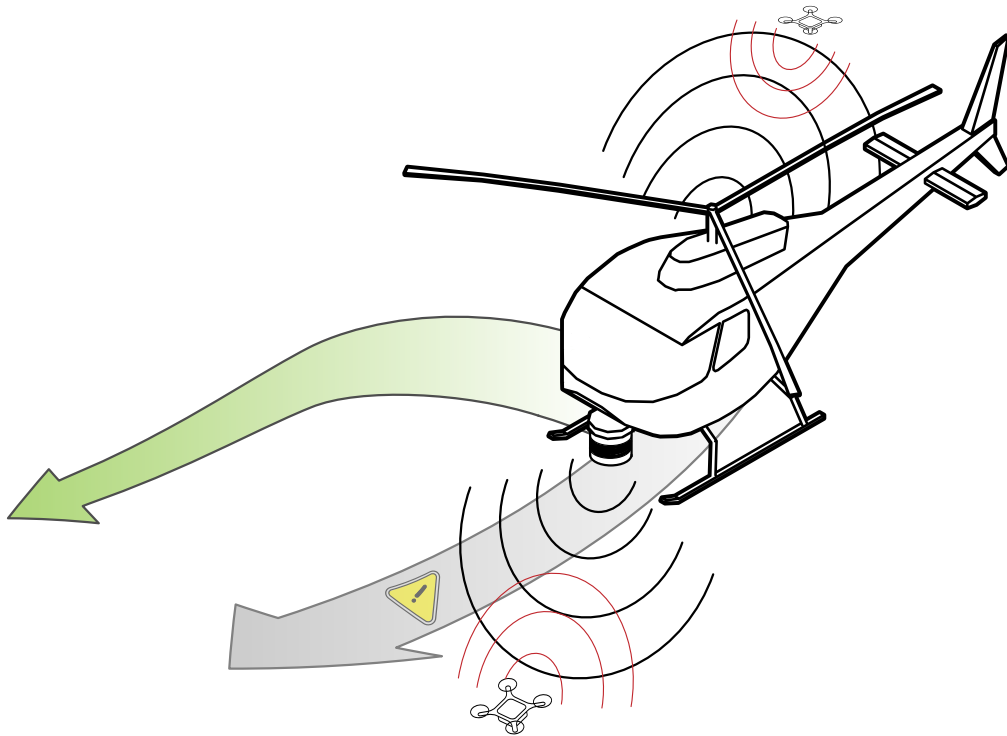


Figure 1.1: Safety-critical encounter between a helicopter and two small UAVs. The helicopter is equipped with a LiDAR sensor to detect and track the intruding drones. Consequently the initial flight path (gray) is replanned to avoid a collision (green).

2. State of the Art

Detect-and-avoid systems aim to enable aircraft or unmanned systems to detect potential conflicts and perform avoidance actions without relying solely on human visual observation. The broader DAA literature distinguishes between cooperative systems, where other traffic broadcasts its state, and non-cooperative systems, where the intruder must be sensed directly. Surveys on sense-and-avoid and UAS collision avoidance highlight that reliable perception, tracking, and conflict assessment remain central challenges for integrating unmanned aircraft into shared airspace fasano2016senseAvoid, skowron2019senseAvoidSmallUAS, pham2015collisionAvoidanceSurvey. For helicopter operations, the problem is especially relevant because helicopters often fly at low altitude and may operate outside controlled airport environments.

2.1 Drone Detection and the Role of LiDAR

To that end drone detection has been studied using multiple sensor modalities, including cameras, radar, radio-frequency sensing, acoustic sensing, thermal imaging, and LiDAR. Recent surveys show that each sensor modality has characteristic strengths and limitations seidalieva2024advancesChallenges, rahman2024uavDetectionSurvey. Camera-based systems are mature and benefit from image-based neural networks, but their performance depends heavily on illumination, weather, contrast, and background appearance. Radar can provide range and velocity estimates, but small drones may have weak radar cross sections, which makes precise localization difficult. LiDAR provides direct geometric range measurements and is therefore attractive for estimating object position in three-dimensional space seidalieva2025lidarUAVDetection.

LiDAR-based detection of small UAVs has been investigated in several works. Hammer et al. studied the potential of LiDAR sensors for UAV detection and later explored detection, tracking, and classification with 360-degree LiDAR systems and sensor fusion hammer2018potentialLidarUAV, hammer2018lidarSmallUAVs, hammer2019sensorFusion360Lidar. Dogru and Marques specifically address drone detection from sparse LiDAR measurements, highlighting the challenge that small aerial objects may generate only a limited number of points dogru2022sparseLidarDrone. More recent work investigates real-time drone detection with airborne LiDAR and deep learning, including synthetic-data-based training and tracking-by-detection pipelines manduhu2023airborneSenseDetect. These works demonstrate the feasibility of

LiDAR-based UAV detection, but they also show that sparsity, range, sensor geometry, and temporal stability remain open issues.

A central limitation in much of the existing work is that detection is often treated as the primary output. For a safety-oriented helicopter application, however, a detector must be embedded into a temporal perception system. A single-frame bounding box does not directly answer whether a drone is approaching, whether it will intersect the helicopter path, or how reliable the estimate is. Therefore, tracking and trajectory prediction are necessary extensions of LiDAR-based drone detection.

2.2 3D LiDAR Object Detection

Deep 3D object detection from point clouds has advanced rapidly in autonomous driving. Early point-based architectures such as PointNet and PointNet++ showed that neural networks can operate directly on unordered point sets qi2017pointnet, qi2017pointnetpp. Voxel-based methods such as VoxelNet and SECOND convert point clouds into structured grids and apply sparse or dense convolutional networks zhou2018voxelnet, yan2018second. PointPillars introduced a particularly efficient representation by grouping points into vertical pillars and producing a bird's-eye-view pseudo-image for fast 2D convolutional detection lang2019pointpillars. Later methods such as PV-RCNN combine voxel and point representations, while CenterPoint formulates 3D detection as center-based object localization and tracking shi2020pvrcnn, yin2021centerpoint. PillarNet further improves pillar-based detection performance while maintaining real-time efficiency shi2022pillarnet.

These architectures are commonly evaluated on large autonomous-driving datasets such as KITTI, nuScenes, and Waymo Open Dataset geiger2012kitti, caesar2020nusenes, sun2020waymo. Such datasets have driven major progress, but their object classes and scene assumptions differ substantially from airborne drone detection. Cars, pedestrians, and cyclists are usually located on or near a ground plane. Their vertical position is therefore highly constrained. In contrast, drones are small airborne objects with free three-dimensional motion. This difference is important because many efficient LiDAR detectors rely heavily on bird's-eye-view representations. They can localize objects accurately in the horizontal plane, but their vertical localization may be less strongly constrained by the architecture and training objective.

For the proposed research, this creates both an opportunity and a challenge. Existing 3D detectors can be adapted to detect drones in LiDAR point clouds, but their outputs must be evaluated not only with generic detection metrics, but also with respect to the quality of temporal state estimation. In particular, errors in height, velocity, and track continuity may be more important for drone DAA than average precision alone.

2.3 Helicopter Detect - and - Avoid and Multi - Object Tracking

Bird Flocking. Drones not so much - gap!

2.4 Research Gap

A practical challenge in LiDAR-based drone perception is the scarcity of suitable datasets. Large public 3D detection datasets focus mostly on autonomous driving, while datasets for small airborne drone detection with LiDAR are limited. Synthetic data can help address this problem by enabling controlled variation of sensor properties, drone geometry, trajectories, ranges, and environmental conditions. However, synthetic data introduces a simulation-to-reality gap. Dieter et al. study this gap for deep-learning-based drone detection and show that synthetic data must be carefully designed and validated to support real-world generalization dieter2023simulationRealityGap. Recent airborne LiDAR work also uses simulation and digital-twin concepts to generate training data for drone detection manduhu2023airborneSenseDetect.

For this dissertation, synthetic data is relevant in two ways. First, it enables systematic generation of sparse LiDAR drone encounters, including rare or safety-critical scenarios that are difficult to collect experimentally. Second, it allows controlled evaluation of tracking and trajectory prediction methods under known ground truth. Nevertheless, the resulting methods must be assessed with respect to generalization. A benchmark should therefore separate training and unseen test scenarios and should report not only detection accuracy, but also tracking continuity, velocity estimation, future position error, and conflict-relevant quantities.

The reviewed literature shows substantial progress in DAA systems, LiDAR-based 3D detection, multi-object tracking, and trajectory prediction. However, the specific problem of tracking and predicting non-cooperative drones from sparse airborne LiDAR remains insufficiently addressed. Existing 3D detectors are largely optimized for ground-based autonomous-driving scenarios. Existing drone detection work often focuses on per-frame detection or ground-based counter-UAV sensing. Existing trajectory prediction methods frequently assume cleaner trajectory observations than are available from sparse LiDAR detections.

The central research gap is therefore the lack of an integrated, safety-oriented pipeline that converts sparse airborne LiDAR measurements of small non-cooperative drones into temporally stable tracks and short-horizon trajectory predictions. Addressing this gap requires combining LiDAR-based detection, uncertainty-aware tracking, and trajectory forecasting in a way that is suitable for helicopter detect-and-avoid scenarios. The proposed dissertation will investigate this connection and evaluate how temporal integration can improve the reliability of drone state estimation beyond single-frame detection.

3. Research Question

3.1 Motivation

The previous chapters have shown that LiDAR-based drone perception is positioned at the intersection of detect-and-avoid, 3D object detection, multi-object tracking, and trajectory prediction. While each of these areas has developed substantially, the specific problem of tracking and predicting the motion of small non-cooperative drones from sparse airborne LiDAR measurements remains insufficiently addressed. Existing detect-and-avoid research emphasizes the need for reliable perception and conflict awareness, but often treats sensing, tracking, and avoidance at a system level [fasano2016senseAvoid](#), [skowron2019senseAvoidSmallUAS](#), [riedel2025daaStandards](#). Existing 3D LiDAR object detection methods, in contrast, have largely been developed for autonomous driving, where objects are typically constrained to the ground plane [lang2019pointpillars](#), [yan2018second](#), [shi2020pvrcnn](#), [yin2021centerpoint](#), [shi2022pillarnet](#).

For a helicopter-oriented detect-and-avoid application, however, the perception problem differs from road-scene perception in several important ways. A drone is small, airborne, and not bound to a road surface. Its vertical position and vertical motion are therefore central parts of its state. Moreover, at safety-relevant distances, a small drone may produce only a small number of LiDAR points, making single-frame detection uncertain or intermittent [dogru2022sparseLidarDrone](#), [seidaliyeva2025lidarUAVDetection](#). A system that only reports independent 3D bounding boxes per frame is therefore insufficient. It must also establish whether detections belong to the same physical object over time, estimate position and velocity, bridge short detection gaps, and predict future motion over a short horizon.

The central premise of this dissertation is that temporal integration is essential for sparse LiDAR-based drone perception. Tracking can reduce the effect of noisy detections, estimate velocity, and provide object identity over time. Trajectory prediction can then use these tracks to estimate future drone positions, which are necessary for downstream conflict assessment. The research gap is therefore not merely the detection of drones in individual LiDAR frames, but the conversion of sparse and uncertain LiDAR detections into temporally stable and prediction-capable intruder state estimates.

3.2 Main Research Question

The main research question of this dissertation is:

How can sparse airborne LiDAR observations of small non-cooperative drones be transformed into robust object tracks and short-horizon trajectory predictions suitable for helicopter detect-and-avoid perception?

This question contains three connected aspects. First, the LiDAR measurements must provide a sufficiently realistic and challenging basis for studying small airborne targets. Second, per-frame detections must be associated and filtered over time to form stable tracks. Third, these tracks must be used to estimate future drone motion over a short time horizon. The dissertation therefore investigates the complete perception chain from sparse LiDAR point clouds to temporally consistent drone state estimation and trajectory forecasting.

3.3 Subquestions

3.3.1 Data and Observation Modeling

The first subquestion concerns the data basis:

Which properties must synthetic and benchmark LiDAR datasets contain in order to evaluate tracking and trajectory prediction of small non-cooperative drones realistically?

This question addresses the fact that suitable public LiDAR datasets for small airborne drone encounters are limited. Existing large-scale datasets such as KITTI, nuScenes, and Waymo Open Dataset are invaluable for autonomous-driving research, but they do not represent the helicopter-oriented drone encounter problem directly [geiger2012kitti](#), [caesar2020nusenes](#), [sun2020waymo](#). Therefore, synthetic data generation and carefully designed benchmark scenarios are necessary. However, synthetic data must be constructed in a way that avoids overly idealized conditions and accounts for the simulation-to-reality gap [dieter2023simulationRealityGap](#), [manduhu2023airborneSenseDetect](#).

This subquestion investigates the role of sensor range, point sparsity, vertical field of view, measurement noise, drone size, drone reflectivity, relative motion, altitude variation, and multi-drone scenarios. In particular, it asks how benchmark data should be structured so that tracking and prediction methods are evaluated on unseen trajectories rather than temporally adjacent frames from the same recording.

3.3.2 Tracking and State Estimation

The second subquestion concerns tracking:

How can noisy and intermittent LiDAR detections of small drones be associated over time to estimate stable position and velocity tracks?

This question focuses on the transition from object detection to object state estimation. The detector provides 3D bounding boxes and confidence scores, but it does not by itself provide persistent identities, velocity estimates, or robustness against missed detections. Tracking-by-detection methods address this by combining motion prediction, detection-to-track association, and recursive state update. Classical Kalman filtering provides a principled framework for estimating object state under noisy measurements [kalman1960newApproach](#). Assignment methods such as the Hungarian algorithm provide a basis for matching detections to tracks in multi-object scenarios [kuhn1955hungarian](#). Practical tracking systems such as SORT and AB3DMOT demonstrate that simple motion models can be effective when combined with reliable detections and appropriate association logic [bewley2016sort](#), [weng2020ab3dmot](#).

For sparse airborne LiDAR drone tracking, this subquestion investigates how such methods must be adapted to small objects, free 3D motion, intermittent detections, and altitude variation. It also asks how detection confidence, point count, and estimated measurement uncertainty can be incorporated into the tracking process.

3.3.3 Short-Horizon Trajectory Prediction

The third subquestion concerns prediction:

How can short-horizon future positions of non-cooperative drones be predicted from LiDAR-based tracks, and how should prediction quality be evaluated for a detect-and-avoid context?

This question extends tracking into future motion forecasting. Simple models such as constant-position, constant-velocity, and Kalman-based prediction provide transparent baselines. More complex learning-based models may capture nonlinear drone motion and maneuver patterns, but they require representative trajectory data and must remain robust to imperfect perception input. Existing work on non-cooperative UAV trajectory prediction and general multi-agent trajectory forecasting provides a foundation, but these methods are often not directly designed for sparse LiDAR-derived tracks [xie2022noncooperativeUAVPrediction](#), [tan2023flyingVehicleTrajectory](#), [salzmann2020trajectronpp](#).

The subquestion therefore investigates both model-based and learning-based predictors. It further asks how prediction performance should be measured. Standard forecasting metrics such as average displacement error and final displacement error are useful, but a helicopter detect-and-avoid application also requires attention to vertical error, missed detections, track continuity, prediction uncertainty, and conflict-relevant quantities such as future separation.

3.4 Scope and Delimitation

The dissertation focuses on perception, tracking, and short-horizon trajectory prediction. It does not aim to develop a complete helicopter flight-control or avoidance system. Instead, the output of the proposed pipeline is intended to provide safety-relevant intruder state estimates that could later be used by a detect-and-avoid or model-predictive-control module. The emphasis is therefore on the reliability of the perception and prediction layer rather than on the final avoidance maneuver.

The work also does not focus solely on improving generic 3D object detection performance. Detection quality remains important, but the dissertation evaluates whether detections can support stable tracks and future trajectory estimates. Consequently, the evaluation will include detection metrics, tracking metrics, prediction metrics, and scenario-level quantities relevant to safety-oriented perception.

4. Methodology

4.1 Overview

The methodology is organized around three main research components that correspond to the identified research gap:

1. construction of a LiDAR drone data and benchmark framework,
2. development of a tracking and state-estimation pipeline from sparse LiDAR detections,
3. development and evaluation of short-horizon trajectory prediction methods.

These components form a pipeline. The first component provides realistic and controllable data. The second component transforms per-frame detections into temporally consistent drone tracks. The third component predicts future drone motion from these tracks. The overall goal is to evaluate how sparse LiDAR observations can be transformed into safety-relevant state and trajectory information for helicopter detect-and-avoid perception.

4.2 Component 1: LiDAR Drone Data and Benchmark Framework

4.2.1 Purpose

The first methodological component establishes the data basis for the dissertation. Since public LiDAR datasets for small airborne drone encounters are limited, a controlled data generation and benchmarking framework is required. The purpose of this component is to generate datasets that contain realistic sparse point-cloud observations, accurate ground-truth object states, and temporally consistent instance annotations.

The benchmark must support three tasks simultaneously: object detection, multi-object tracking, and trajectory prediction. Therefore, the data must include not only point clouds and 3D bounding boxes, but also object identities, timestamps, scene structure, and temporally linked annotations.

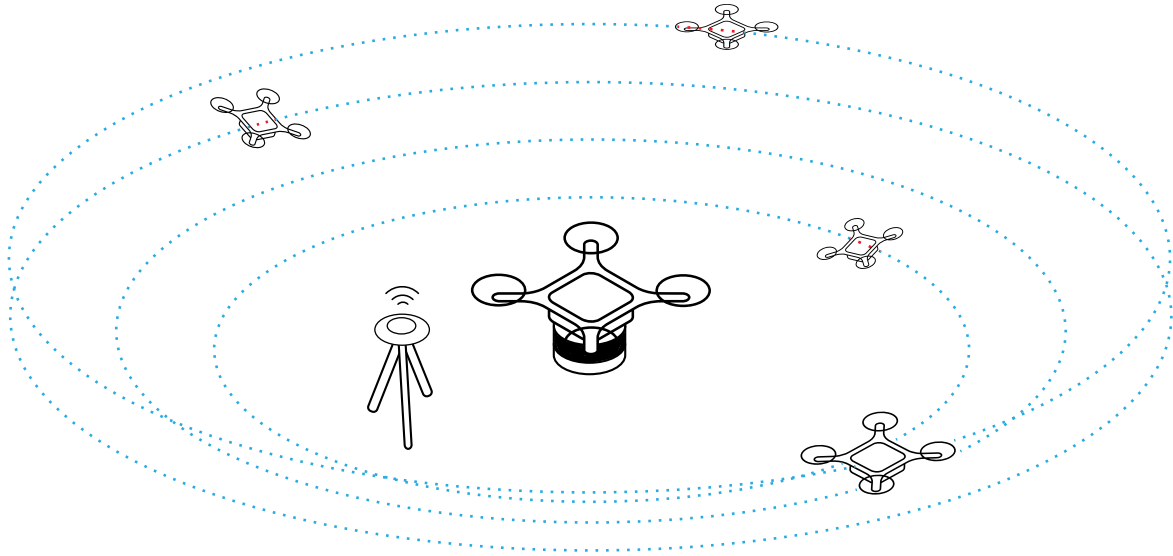


Figure 4.1: Dummy caption describing the LiDAR drone data generation framework.

4.2.2 Synthetic LiDAR Data Generation

A simulation environment will be used to generate LiDAR observations of drone encounters. The sensor model will include parameters such as maximum range, vertical field of view, angular resolution, range noise, dropout, and intensity variation. The purpose is not to create visually realistic scenes alone, but to create point-cloud statistics that are useful for studying sparse airborne object perception.

The generated scenarios will vary the number of drones, distance to the sensor, relative altitude, crossing angle, approach speed, and trajectory type. Special attention will be given to altitude variation because flying drones are not constrained to the ground plane. This is important because many efficient 3D detectors use bird's-eye-view or pillar-based representations, which can be strong in horizontal localization but weaker in vertical localization [lang2019pointpillars](#), [shi2022pillarnet](#). The dataset will therefore deliberately include scenarios where drones occupy similar horizontal positions but different heights.

The data generation process will also include negative and hard-negative examples. Empty scenes, background clutter, distant drones with few returns, and partial visibility cases are necessary to evaluate false positives and robustness. This is important because a detect-and-avoid perception system must be reliable not only in ideal target-present cases, but also under uncertainty and absence of targets.

4.2.3 Dataset Format and Splitting

The generated data will be organized in a nuScenes-like format because this structure is supported by common LiDAR detection toolchains and allows temporal linking of samples, sample data, ego poses, and annotations [caesar2020nusenes](#), [openpcdet2020](#). Each frame will contain a LiDAR point cloud, sensor pose, 3D object annotations, instance identifiers, and timestamps. Annotation chains will be preserved so that drone trajectories can be reconstructed from the dataset.

The dataset will be split at the level of runs or scenarios rather than individual frames. This prevents temporally adjacent frames from appearing in both training and test data. The intended split consists of:

- a training set for model fitting,
- a validation set for parameter selection,
- an unseen benchmark set for final performance assessment.

This scenario-level separation is essential because trajectory prediction and tracking must be evaluated on unseen motion patterns, not merely on neighboring frames from the same sequence.

4.2.4 Baseline Detection Models

The first experimental baseline will use established LiDAR 3D object detection architectures. Pillar-based models such as PointPillars and PillarNet are particularly relevant because of their computational efficiency and suitability for real-time applications [lang2019pointpillars](#), [shi2022pillarnet](#). Voxel- and point-based alternatives such as SECOND, PV-RCNN, and CenterPoint will be considered as comparison points because they represent different trade-offs between computational efficiency and 3D geometric expressiveness [yan2018second](#), [shi2020pvrcnn](#), [yin2021centerpoint](#).

Detection performance will be evaluated using standard metrics such as precision, recall, and average precision. In addition, the dissertation will report center-position error, horizontal error, vertical error, size error, and detection performance as a function of distance and point count. These additional metrics are necessary because average precision alone does not show whether detections are accurate enough for tracking and trajectory prediction.

4.3 Component 2: Tracking and State Estimation from Sparse LiDAR Detections

4.3.1 Purpose

The second methodological component transforms per-frame detections into temporally stable object tracks. This is necessary because single-frame detections can be noisy, may disappear for short periods, and do not directly provide persistent object identity or velocity. A tracking module estimates the state of each drone over time and provides the input for trajectory prediction.

4.3.2 Tracking-by-Detection Pipeline

The tracking pipeline follows the tracking-by-detection paradigm. In each frame, the detector produces a set of 3D bounding boxes and confidence scores. The center of each predicted box is interpreted as a noisy measurement of a drone position. Existing tracks are predicted forward to the current timestamp, detections are assigned to tracks, and matched tracks are updated with the new measurements.

The first baseline will use nearest-neighbor association and a constant-velocity motion model. This provides a transparent reference implementation and helps identify the effect of detection noise and missed detections. The second baseline will use a Kalman filter for recursive state estimation [kalman1960newApproach](#). For each track, the state vector is defined as $\mathbf{x}_t = [p_x \ p_y \ p_z \ v_x \ v_y \ v_z]^T$,

where (p_x, p_y, p_z) denote three-dimensional position and (v_x, v_y, v_z) denote velocity. A constant-velocity transition model predicts the next state, while the detector provides noisy position measurements. The Kalman filter then combines prediction and measurement correction to obtain a smoothed state estimate.

For multi-drone scenarios, detection-to-track association will first be implemented with distance-based nearest-neighbor matching. It will then be extended to an assignment formulation using the Hungarian algorithm [kuhn1955hungarian](#). This allows simultaneous association of multiple detections and tracks. The tracking design is inspired by efficient tracking-by-detection systems such as SORT and AB3DMOT, while adapting the state and measurement assumptions to sparse airborne LiDAR drone observations [bewley2016sort](#), [weng2020ab3dmot](#).

4.3.3 Uncertainty Handling

A key methodological element is uncertainty handling. Sparse LiDAR detections of small drones can vary strongly in quality depending on range, viewing angle, and number of returns. Therefore, the tracking module will investigate whether measurement uncertainty can be adapted using detection confidence, number of LiDAR points inside the detection, or estimated spatial spread of the observed points.

In a Kalman-filter setting, this can be represented through the measurement covariance. A high-confidence detection with many supporting points can be treated as a more reliable measurement, while a low-confidence detection or a detection with very few points can be assigned higher measurement uncertainty. This allows the tracker to rely more strongly on the motion model when detections are weak, and more strongly on measurements when detections are reliable.

Missed detections will be handled by prediction-only updates for a limited number of frames. If no detection is associated with a track, the state will be propagated using the motion model and the uncertainty will increase. After a configurable number of missed frames, the track will be terminated. This mechanism is important because small drones may disappear temporarily due to sparse point returns, but tracks should not be kept indefinitely without evidence.

4.3.4 Tracking Evaluation

Tracking will be evaluated both independently and as input to prediction. The metrics will include:

- track recall and track precision,
- number of identity switches,
- average track duration,
- position error over time,
- velocity estimation error,
- robustness to missed detections,
- runtime per frame.

The evaluation will compare raw detector centers, nearest-neighbor tracking, Kalman-filter tracking, and assignment-based multi-object tracking. The purpose is to determine whether temporal filtering improves state estimation sufficiently to support trajectory prediction.

4.4 Component 3: Short-Horizon Trajectory Prediction

4.4.1 Purpose

The third methodological component predicts future drone positions from the estimated tracks. In a helicopter detect-and-avoid context, short-horizon prediction is necessary because avoidance decisions depend not only on where an intruder is now, but also on where it is likely to be in the near future. The dissertation therefore investigates how reliable future trajectory estimates can be produced from sparse LiDAR-derived tracks.

4.4.2 Model-Based Prediction Baselines

The first prediction methods will be simple model-based baselines. These include constant-position, constant-velocity, and Kalman-based prediction. A constant-velocity predictor estimates future position according to $[pt + vt\Delta t.]$

Here, (pt) and (vt) are provided by the tracking module. The Kalman-based variant additionally propagates the state covariance, which provides an estimate of prediction uncertainty. These baselines are computationally efficient, interpretable, and suitable for short prediction horizons. They also provide a necessary reference point for evaluating whether more complex learned models are justified.

4.4.3 Learning-Based Prediction Models

In a second step, learning-based predictors will be investigated. These models receive a sequence of past drone states and predict a sequence of future positions. The input may consist of positions only, or it may include velocities, detection confidence, point-count information, and track uncertainty. The first learning-based baselines will use simple sequence models trained on ground-truth trajectories. Later experiments will use tracks generated by the detection and tracking pipeline to evaluate robustness under realistic perception errors.

Existing work on non-cooperative UAV trajectory prediction shows that future motion can be predicted online from observed flight behavior xie2022noncooperativeUAVPrediction. General multi-agent trajectory prediction methods such as Social-LSTM and Trajectron++ demonstrate how temporal history and interaction structure can be incorporated into learned forecasting models alahi2016socialLSTM, salzmann2020trajectronpp. In this dissertation, such ideas will be adapted cautiously, with a focus on short-horizon prediction and sparse LiDAR-derived input rather than idealized ground-truth trajectories.

A central comparison will be made between predictors trained on clean ground-truth tracks and predictors evaluated on noisy detector-based tracks. This distinction is essential because a model that performs well on perfect trajectories may fail when the input contains missed detections, identity switches, or noisy height estimates.

4.4.4 Prediction Uncertainty

For safety-relevant perception, a single deterministic future trajectory may be insufficient. A drone may maneuver unexpectedly, and the perception system may be uncertain about its current state. Therefore, the methodology will include uncertainty-aware prediction. For Kalman-based methods, uncertainty is obtained through covariance propagation. For learning-based models, uncertainty may be estimated through probabilistic outputs, model ensembles, or sampling-based prediction.

The purpose of uncertainty estimation is not to solve the final avoidance problem, but to provide a richer input to a future conflict assessment module. A predicted position with high uncertainty should be interpreted differently from a predicted position with low uncertainty. This distinction is important for safety-oriented decision making.

4.4.5 Trajectory Prediction Evaluation

Prediction quality will be evaluated using standard forecasting metrics and application-specific metrics. The standard metrics include:

- average displacement error,
- final displacement error,
- horizontal displacement error,
- vertical displacement error,
- error as a function of prediction horizon.

In addition, the dissertation will evaluate how prediction quality depends on detection quality, track length, point sparsity, drone distance, and missed detections. For helicopter detect-and-avoid relevance, scenario-level quantities such as future separation and time-to-conflict-oriented indicators will be considered. These metrics will not replace a full avoidance system, but they will indicate whether the perception pipeline produces information that is useful for downstream safety logic.

4.5 Experimental Procedure

The experimental procedure will follow a staged design. First, synthetic LiDAR datasets will be generated and organized into training, validation, and benchmark splits. Second, baseline 3D detectors will be trained and evaluated. Third, tracking methods will be applied to detector outputs and compared against ground-truth trajectories. Fourth, trajectory prediction methods will be trained and evaluated using both ideal ground-truth tracks and realistic detector-based tracks. Finally, the complete pipeline will be evaluated on unseen benchmark scenarios.

The experiments will therefore separate four levels of performance:

1. single-frame detection performance,
2. tracking and state-estimation performance,
3. trajectory prediction performance from ground-truth tracks,
4. trajectory prediction performance from detector-based tracks.

This separation is important because errors can propagate through the pipeline. A poor prediction may be caused by an unsuitable prediction model, but it may also be caused by detector noise, wrong data association, height errors, or missed detections. The staged evaluation allows these effects to be analyzed separately.

4.6 Expected Contribution

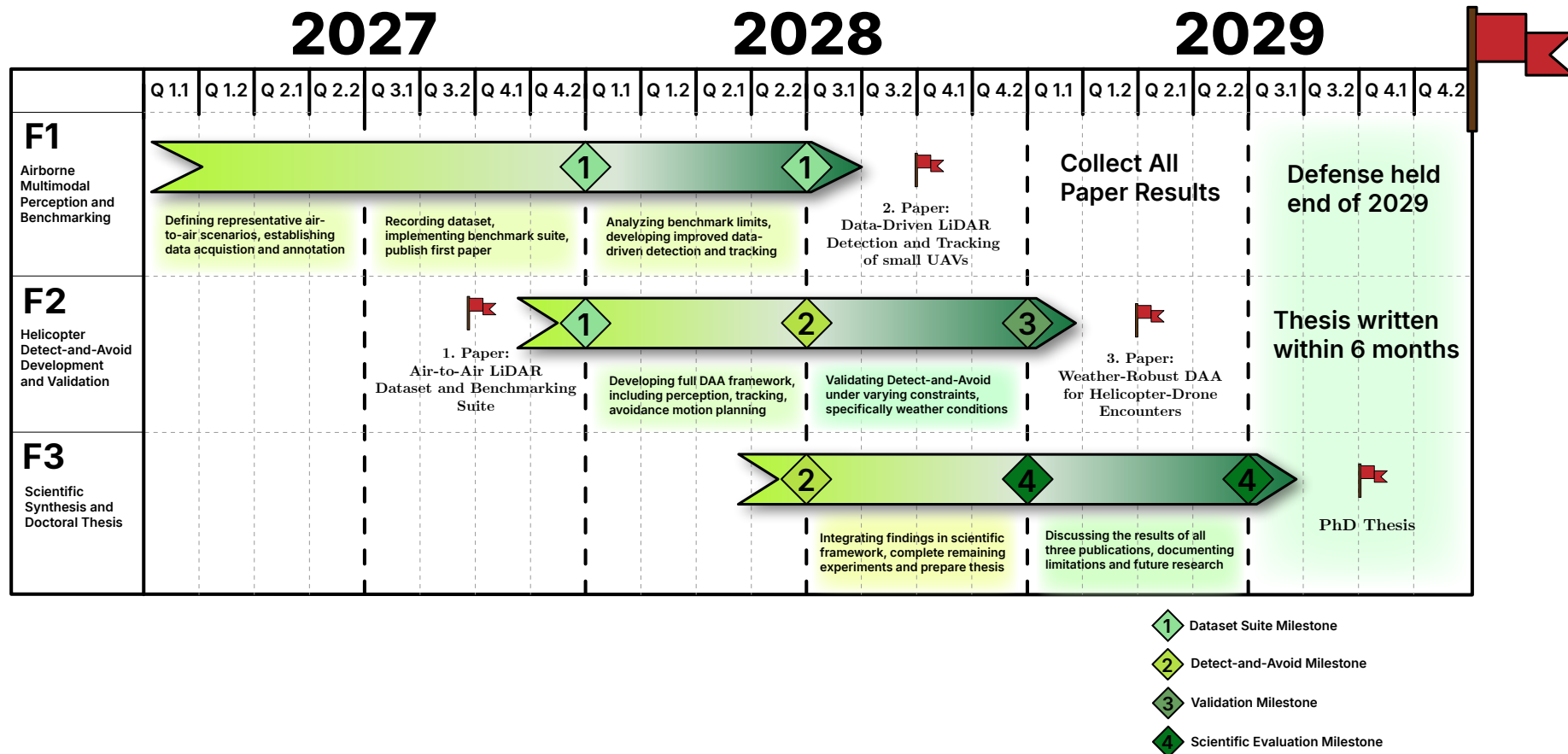
The expected contribution of the methodology is an integrated framework for sparse LiDAR-based drone tracking and trajectory prediction. The dissertation aims to contribute:

- a benchmark structure for small airborne drone encounters in LiDAR point clouds,
- a systematic analysis of how sparse detections affect tracking and prediction,
- a tracking pipeline for non-cooperative drones using LiDAR-based detections,
- a comparison of model-based and learning-based short-horizon trajectory predictors,
- an evaluation methodology that connects detection, tracking, and prediction performance to helicopter detect-and-avoid perception requirements.

The scientific value lies in bridging the gap between per-frame LiDAR object detection and safety-relevant temporal perception. Instead of treating drone detection as an isolated bounding-box prediction problem, the proposed methodology studies how sparse airborne LiDAR observations can support continuous intruder state estimation and future motion prediction.

Chapter 5

Research Plan



6. Publication Plan

This chapter brings together the central insights from our work on training and evaluating Reinforcement Learning (RL) agents for decentralized multi-quadrotor cable-suspended payload transport. We revisit our research objectives and contributions, summarize the most important findings, consider their broader significance, acknowledge the limitations we encountered, and suggest directions for future research to further advance the field.

6.1 Key Findings

In pursuit of our research objectives, we first formulated the decentralized multi Unmanned Aerial Vehicle (UAV) payload transport problem as a multi agent Decentralized Partially Observable Markov Decision Process (Dec-POMDP) with local observation and action spaces enabling coordination through decentralized policies, then designed and implemented a high-performance GPU-parallel simulation pipeline to support scalable policy learning, investigated reward design by decomposing the overall objective into tracking and stability components, evaluated the learned policies under harsh initializations to assess resilience relative to classical baselines, and finally demonstrated that our decentralized framework maintains precise tracking and payload stability as the number of UAVs increases.

Concurrently, running training across thousands of parallel simulation environments effectively reduced training time and gradient variance, accelerating convergence to reliable policy performance. Our proposed training pipeline, implemented in JAX and MuJoCo, enables very capable and efficient training.

Randomizing the initial positions of payloads and quadrotors during training was crucial to train a policy to recover from harsh initial conditions. We found that applying domain randomization solely to the maximum thrust parameter, without randomizing parameters such as payload mass, quadrotor mass, and cable length, still leads to decent generalization in simulation. Extending each agent’s observation to include the most recent action further enhanced the stability and smoothness of the control outputs.

The design of the reward function proved to be another key factor. By shaping reward components with exponential decay rather than fixed penalties, we prevented agents from terminating episodes prematurely and enabled them to

recover gracefully from disturbances. Our experiments demonstrate that including a regularization term on the agent's actions is critical to producing smooth control signals and avoiding abrupt bang-bang behavior. This regularization also improves the ability to transfer policies from simulation to real hardware.

Notably, policies trained exclusively on recovery and position holding transferred successfully to dynamic trajectory tracking. This result highlights the flexibility of the learned control strategies. Despite these advances, achieving a reliable sim-to-real transfer remains an open challenge.

6.2 Implications

Our findings underscore the potential of RL for cooperative aerial transport tasks that are difficult to handle with traditional model-based control. The decentralized Multi-Agent Reinforcement Learning (MARL) framework we developed is capable of stabilizing suspended payloads, following precise trajectories, and rejecting external disturbances. Such capabilities are directly relevant to real-world applications, including aerial construction support, infrastructure inspection, and rapid delivery of emergency supplies.

At the same time, our results emphasize that accurate and robust state estimation is essential for any practical deployment. Our approach requires the position information of all quadrotors to be shared with low latency and high frequency. In many real-world environments, full state information is not available, external motion capture systems are impractical, and low-latency communication is challenging. Ensuring that learned policies remain resilient in the face of communication issues, sensor noise, and estimation errors is therefore a critical requirement for future work.

6.3 Limitations

Several important limitations must be mentioned in the conclusions of this study. First, learned policies sometimes fail under extreme conditions, for example, when payload oscillations grow large or quadrotor separation becomes excessive. Our current approach does not include obstacle avoidance or integrated motion planning and relies heavily on external infrastructure without providing formal safety guarantees. Each change in payload mass or quadrotor configuration presently requires retraining from scratch, which limits flexibility. Although we observed some generalization across payload masses and cable lengths in simulation, systematic validation on real platforms is still needed.

Moreover, our methods depend on knowing quantities such as payload mass and cable lengths in advance. Improving autonomy will require onboard estimation of these variables. Addressing the challenge of accurately estimating the full system parameters of cables and payload remains an important barrier to fully autonomous operation.

As mentioned above, our method requires high-frequency position updates from all quadrotors, which is not always feasible in real-world scenarios. The current approach also does not account for communication delays or packet loss, which can significantly impact performance in practice. Future work should explore ways to make the system more robust to these issues, for example, by incorporating predictive control or state estimation techniques, or by estimating the positions of other quadrotors solely from local observations of each quadrotor.

Finally, while our training pipeline is highly efficient, it still requires substantial computational resources for training. Future work should explore more sample-efficient algorithms and methods to reduce the reliance on extensive simulation data.

6.4 Future Research Directions

To address these challenges and enhance the real-world applicability of decentralized RL for multi-quadrotor cable-suspended payload transport, we propose several avenues for future research. Enriching the agent’s state representation with extended histories or estimates of latent variables such as cable length, payload mass, quadrotor mass, and relative positions could yield more robust control. Developing onboard estimation schemes that leverage Inertial Measurement Unit (IMU) cameras or LiDAR sensors would remove dependence on external motion capture systems. These should be combined with motion and path planning algorithms to ensure safe operation in dynamic environments. To facilitate real-world deployment, inter-quadrotor communication must be investigated, including methods for handling communication delays and packet loss.

Expanding experiments to a wider range of payload shapes, masses, and cable configurations will help characterize the limits of scalability and generalization. Improving sim-to-real transfer by applying advanced domain randomization methods may reduce the remaining reality gap. Finally, incorporating online policy adaptation mechanisms and exploring alternative RL algorithms, such as off-policy actor-critic methods, recurrent networks, attention mechanisms, or meta-learning approaches, could accelerate training and improve policy flexibility under changing conditions.

Pursuing these directions will bring us closer to realizing fully autonomous, reliable, and efficient multi-agent cable-suspended aerial transport systems based on RL.

7. Abstract

Name: Max Klemens

Status: PhD Inception, first year

Supervisor: Prof. Dr. Christian Seidel

Multimodal Airborne Perception and Detect-and-Avoid for Non-Cooperative Drones

The increasing operation of unmanned aerial vehicles in low-altitude airspace introduces additional risks for crewed aviation, particularly when small drones operate without cooperative identification or communication systems. Conventional detect-and-avoid approaches rely on ground-based sensors or cooperative information sources such as transponders. These approaches are insufficient for the reliable detection of small, non-cooperative drones from a moving airborne platform. The objective of this doctoral research is therefore to investigate multimodal airborne perception and detect-and-avoid methods for helicopter encounters with non-cooperative unmanned aerial vehicles.

The research is structured into three consecutive scientific contributions. The first contribution addresses the development of a synchronized air-to-air RGB-LiDAR dataset and benchmarking framework for the detection, localization, and tracking of small aerial targets. The dataset is intended to include real and simulated sensor data. Particular emphasis will be placed on varying target count (up to six target drones), backgrounds and environmental conditions.

The second contribution will investigate data-driven methods for LiDAR-based detection and temporal tracking of small aerial objects. Existing three-dimensional detection methods are primarily designed for road-based environments and commonly assume objects located close to a ground plane. These assumptions are not directly transferable to unrestricted airspace, where targets may occur at arbitrary elevations and may be represented by only a small number of LiDAR measurements. The planned work will therefore examine suitable spatial representations, temporal point-cloud integration, motion information, and tracking strategies for sparse aerial targets.

The third contribution will integrate the developed perception and tracking methods into a helicopter-oriented detect-and-avoid framework. Representative encounter scenarios will be evaluated under different trajectories, visibility conditions, weather conditions, and sensor degradations.

The research is currently in the requirements-definition phase. Initial work includes the integration of LiDAR sensing, optical reference tracking, simulation, data recording, and annotation components.

Feedback is specifically requested regarding the scientific delimitation of the three contributions, and the novelty and scope of the proposed airborne benchmark.

Bibliography

- [1] M. Idrissi, M. Salami, and F. Annaz, “A review of quadrotor unmanned aerial vehicles: Applications, architectural design and control algorithms,” *Journal of Intelligent & Robotic Systems*, vol. 104, pp. 1–33, 2022.
- [2] M. Lyu, Y. Zhao, C. Huang, and H. Huang, “Unmanned aerial vehicles for search and rescue: A survey,” *Remote. Sens.*, vol. 15, p. 3266, 2023.
- [3] J. Chen, J. Sun, and G. Wang, “From unmanned systems to autonomous intelligent systems,” *Engineering*, 2021.
- [4] P. Abbaraju, R. Voyles, et al., “Sensing and sampling of trace contaminations by a dexterous hexrotor uav at nuclear facilities-18600,” in *Proceedings of the WM2018 Symposium Conference, Phoenix, AZ, USA*, 2018, pp. 18–22.
- [5] S. Batra, Z. Huang, A. Petrenko, T. Kumar, A. Molchanov, and G. S. Sukhatme, “Decentralized control of quadrotor swarms with end-to-end deep reinforcement learning,” in *Proceedings of the 5th Conference on Robot Learning*, Jan. 11, 2022, pp. 576–586. DOI: 10.48550/arxiv.2109.07735 eprint: 2109.07735.
- [6] X. Zhou, J. Zhu, H. Zhou, C. Xu, and F. Gao, “Ego-swarm: A fully autonomous and decentralized quadrotor swarm system in cluttered environments,” *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4101–4107, 2020.
- [7] J. Estevez, G. Garate, J. M. Lopez-Guede, and M. Larrea, “Review of aerial transportation of suspended-cable payloads with quadrotors,” *Drones*, vol. 8, p. 35, Feb. 2024, ISSN: 2504-446X. DOI: 10.3390/drones8020035
- [8] J. Hwangbo, I. Sa, R. Siegwart, and M. Hutter, “Control of a quadrotor with reinforcement learning,” *IEEE Robotics and Automation Letters*, vol. 2, pp. 2096–2103, 2017. DOI: 10.48550/arxiv.1707.05110 eprint: 1707.05110.
- [9] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, and D. Scaramuzza, “Champion-level drone racing using deep reinforcement learning,” *Nature*, vol. 620, pp. 982–987, Aug. 2023, ISSN: 1476-4687. DOI: 10.1038/s41586-023-06419-4
- [10] Y. Song, A. Romero, M. Müller, V. Koltun, and D. Scaramuzza, “Reaching the limit in autonomous racing: Optimal control versus reinforcement learning,” *Science Robotics*, vol. 8, 2023.
- [11] Z. Huang, Z. Yang, R. Krupani, B. Şenbaşlar, S. Batra, and G. S. Sukhatme, “Collision avoidance and navigation for a quadrotor swarm using end-to-end deep reinforcement learning,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, May 2024, pp. 300–306. DOI: 10.1109/ICRA57147.2024.10611499
- [12] J. Eschmann, D. Albani, and G. Loianno, “Learning to fly in seconds,” *IEEE Robotics and Automation Letters*, vol. 9, no. 7, pp. 6336–6343, Jul. 2024, ISSN: 2377-3766. DOI: 10.1109/LRA.2024.3396025

-
- [13] K. Wahba, J. Ortiz-Haro, M. Toussaint, and W. Hönig, “Kinodynamic motion planning for a team of multirotors transporting a cable-suspended payload in cluttered environments,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2024, pp. 12 750–12 757. DOI: 10.1109/IR0S58592.2024.10802794
 - [14] J. Chen et al., “What matters in learning a zero-shot sim-to-real rl policy for quadrotor control? a comprehensive study,” *IEEE Robotics and Automation Letters*, vol. 10, no. 7, pp. 7134–7141, Jul. 2025, ISSN: 2377-3774. DOI: 10.1109/lra.2025.3575011
 - [15] M. L. Littman, “Markov games as a framework for multi-agent reinforcement learning,” in *Machine learning proceedings 1994*, 1994, pp. 157–163.
 - [16] K. Sreenath and V. Kumar, “Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadrotor robots,” in *Robotics: Science and Systems IX*, Jun. 23, 2013, ISBN: 978-981-07-3937-9. DOI: 10.15607/RSS.2013.IX.011
 - [17] D. K. D. Villa, A. S. Brandão, R. Carelli, and M. Sarcinelli-Filho, “Cooperative load transportation with two quadrotors using adaptive control,” *IEEE Access*, vol. 9, pp. 129 148–129 160, 2021, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2021.3113466
 - [18] T. Lee, “Geometric control of quadrotor UAVs transporting a cable-suspended rigid body,” *IEEE Transactions on Control Systems Technology*, vol. 26, pp. 255–264, Jan. 2018, ISSN: 1558-0865. DOI: 10.1109/TCST.2017.2656060
 - [19] M. Tognon, C. Gabellieri, L. Pallottino, and A. Franchi, “Aerial co-manipulation with cables: The role of internal force for equilibria, stability, and passivity,” *IEEE Robotics and Automation Letters*, vol. 3, pp. 2577–2583, Jul. 2018, ISSN: 2377-3766. DOI: 10.1109/LRA.2018.2803811
 - [20] A. Tagliabue, M. Kamel, R. Siegwart, and J. I. Nieto, “Robust collaborative object transportation using multiple mavs,” *Int. J. Robotics Res.*, vol. 38, no. 9, 2019. DOI: 10.1177/0278364919854131
 - [21] S. Sun and A. Franchi, “Nonlinear MPC for full-pose manipulation of a cable-suspended load using multiple UAVs,” in *2023 International Conference on Unmanned Aircraft Systems (ICUAS)*, Jun. 2023, pp. 969–975. DOI: 10.1109/ICUAS57906.2023.10156031
 - [22] G. Li and G. Loianno, “Nonlinear model predictive control for cooperative transportation and manipulation of cable suspended payloads with multiple quadrotors,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2023, pp. 5034–5041. DOI: 10.1109/IR0S55552.2023.10341785 eprint: 2303.06165.
 - [23] B. E. Jackson, T. A. Howell, K. Shah, M. Schwager, and Z. Manchester, “Scalable cooperative transport of cable-suspended loads with UAVs using distributed trajectory optimization,” *IEEE Robotics and Automation Letters*, vol. 5, pp. 3368–3374, Apr. 2020, ISSN: 2377-3766. DOI: 10.1109/LRA.2020.2975956
 - [24] C. Gabellieri, M. Tognon, D. Sanalidro, and A. Franchi, “Equilibria, stability, and sensitivity for the aerial suspended beam robotic system subject to parameter uncertainty,” *IEEE Transactions on Robotics*, vol. 39, pp. 3977–3993, Oct. 2023, ISSN: 1941-0468. DOI: 10.1109/TR0.2023.3279033 eprint: 2302.07031.
 - [25] Y. Wang, J. Wang, X. Zhou, T. Yang, C. Xu, and F. Gao, “Safe and agile transportation of cable-suspended payload via multiple aerial robots,” *CoRR*, vol. abs/2501.15272, 2025. DOI: 10.48550/ARXIV.2501.15272 arXiv: 2501.15272.
-

-
- [26] W. Koch, R. Mancuso, R. West, and A. Bestavros, “Reinforcement learning for uav attitude control,” *ACM Transactions on Cyber-Physical Systems*, vol. 3, pp. 1–21, 2018. DOI: 10.48550/arxiv.1804.04154 eprint: 1804.04154.
 - [27] T. P. Lillicrap et al., “Continuous control with deep reinforcement learning,” *CoRR*, vol. abs/1509.02971, 2015. DOI: 10.48550/arxiv.1509.02971 eprint: 1509.02971.
 - [28] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, “Trust region policy optimization,” *CoRR*, vol. abs/1502.05477, 2015. DOI: 10.48550/arxiv.1502.05477 arXiv: 1502.05477. [Online]. Available: <http://arxiv.org/abs/1502.05477>
 - [29] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *CoRR*, vol. abs/1707.06347, 2017. DOI: 10.48550/arxiv.1707.06347 arXiv: 1707.06347. [Online]. Available: <http://arxiv.org/abs/1707.06347>
 - [30] J. Xing, I. Geles, Y. Song, E. Aljalbout, and D. Scaramuzza, “Multi-task reinforcement learning for quadrotors,” *IEEE Robotics and Automation Letters*, pp. 1–8, 2024, ISSN: 2377-3766. DOI: 10.1109/LRA.2024.3520894
 - [31] A. Molchanov, T. Chen, W. Hönig, J. A. Preiss, N. Ayanian, and G. S. Sukhatme, “Sim-to-(multi)-real: Transfer of low-level robust control policies to multiple quadrotors,” in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Nov. 2019, pp. 59–66. DOI: 10.1109/IROS40897.2019.8967695
 - [32] E. Kaufmann, L. Bauersfeld, and D. Scaramuzza, “A benchmark comparison of learned control policies for agile quadrotor flight,” in *2022 International Conference on Robotics and Automation (ICRA)*, May 2022, pp. 10 504–10 510. DOI: 10.1109/ICRA46639.2022.9811564
 - [33] S. Gronauer, M. Kissel, L. Sacchetto, M. Korte, and K. Diepold, “Using simulation optimization to improve zero-shot policy transfer of quadrotors,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2022, pp. 10 170–10 176. DOI: 10.1109/IROS47612.2022.9981229 eprint: 2201.01369.
 - [34] H. Ma, Z. Ren, B. Dai, and N. Li, “Skill transfer and discovery for sim-to-real learning: A representation-based viewpoint,” *technical report*, 2024. DOI: 10.48550/arxiv.2404.05051 eprint: 2404.05051.
 - [35] Z. Huang et al., “Quadswarm: A modular multi-quadrotor simulator for deep reinforcement learning with direct thrust control,” *CoRR*, vol. abs/2306.09537, 2023. DOI: 10.48550/ARXIV.2306.09537 arXiv: 2306.09537.
 - [36] S. Belkhale, R. Li, G. Kahn, R. McAllister, R. Calandra, and S. Levine, “Model-based meta-reinforcement learning for flight with suspended payloads,” *IEEE Robotics and Automation Letters*, vol. 6, pp. 1471–1478, Apr. 2021, ISSN: 2377-3766, 2377-3774. DOI: 10.1109/LRA.2021.3057046 eprint: 2004.11345.
 - [37] H. Hua, Y. Fang, X. Zhang, and C. Qian, “A new nonlinear control strategy embedded with reinforcement learning for a multirotor transporting a suspended payload,” *IEEE/ASME Transactions on Mechatronics*, vol. 27, pp. 1174–1184, Apr. 2022, ISSN: 1941-014X. DOI: 10.1109/TMECH.2021.3082897
 - [38] C. A. Dimmig et al., “Survey of simulators for aerial robots: An overview and in-depth systematic comparisons [survey],” *IEEE Robotics Autom. Mag.*, vol. 32, no. 2, pp. 153–166, 2025. DOI: 10.1109/MRA.2024.3433171 arXiv: 2311.02296.
 - [39] G. Li, X. Liu, and G. Loianno, “Rotortm: A flexible simulator for aerial transportation and manipulation,” *IEEE Transactions on Robotics*, vol. 40, pp. 831–850, 2022.
-

-
- [40] M. Kulkarni, W. Rehberg, and K. Alexis, "Aerial gym simulator: A framework for highly parallelized simulation of aerial robots," *IEEE Robotics and Automation Letters*, vol. 10, pp. 4093–4100, 2025. DOI: 10.1109/LRA.2025.3548507
 - [41] B. Xu, F. Gao, C. Yu, R. Zhang, Y. Wu, and Y. Wang, "OmniDrones: An efficient and flexible platform for reinforcement learning in drone control," *IEEE Robotics and Automation Letters*, vol. 9, pp. 2838–2844, Mar. 2024, ISSN: 2377-3766. DOI: 10.1109/LRA.2024.3356168
 - [42] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," *CoRR*, vol. abs/1706.02275, 2017. DOI: 10.48550/arxiv.1706.02275 arXiv: 1706.02275. [Online]. Available: <http://arxiv.org/abs/1706.02275>
 - [43] C. Yu et al., "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., 2022. [Online]. Available: http://papers.nips.cc/paper_files/paper/2022/hash/9c1535a02f0ce079433344e14d910597-Abstract-Datasets_and_Benchmarks.html
 - [44] C. S. de Witt et al., "Is independent learning all you need in the starcraft multi-agent challenge?" *CoRR*, vol. abs/2011.09533, 2020. DOI: 10.48550/arxiv.2011.09533 arXiv: 2011.09533. [Online]. Available: <https://arxiv.org/abs/2011.09533>
 - [45] G. Papoudakis, F. Christianos, L. Schäfer, and S. V. Albrecht, "Benchmarking multi-agent deep reinforcement learning algorithms in cooperative tasks," in *NeurIPS Datasets and Benchmarks*, 2020.
 - [46] Y. Chen et al., "Towards human-level bimanual dexterous manipulation with reinforcement learning," *CoRR*, vol. abs/2206.08686, 2022. DOI: 10.48550/ARXIV.2206.08686 arXiv: 2206.08686.
 - [47] B. Pandit et al., "Learning decentralized multi-biped control for payload transport," in *Conference on Robot Learning*, 2024. DOI: 10.48550/arxiv.2406.17279 eprint: 2406.17279.
 - [48] Y. Ji, B. Zhang, and K. Sreenath, "Reinforcement learning for collaborative quadrupedal manipulation of a payload over challenging terrain," *2021 IEEE 17th International Conference on Automation Science and Engineering (CASE)*, pp. 899–904, 2021. DOI: 10.1109/case49439.2021.9551481
 - [49] W. Chen, M. Nguyen, Z. Li, G. N. Sue, and K. Sreenath, "Decentralized navigation of a cable-towed load using quadrupedal robot team via MARL," *CoRR*, vol. abs/2503.18221, 2025. DOI: 10.48550/ARXIV.2503.18221 arXiv: 2503.18221.
 - [50] Y. Xie et al., "Multi-uav behavior-based formation with static and dynamic obstacles avoidance via reinforcement learning," *CoRR*, vol. abs/2410.18495, 2024. DOI: 10.48550/ARXIV.2410.18495 arXiv: 2410.18495.
 - [51] X. Diao and J. Wang, "Efficient multi-agent navigation with lightweight DRL policy," *CoRR*, vol. abs/2408.16370, 2024. DOI: 10.48550/ARXIV.2408.16370 arXiv: 2408.16370.
 - [52] Z. Zhao, Y. Wan, and Y. Chen, "Deep reinforcement learning-driven collaborative rounding-up for multiple unmanned aerial vehicles in obstacle environments," *Drones*, vol. 8, p. 464, Sep. 2024, ISSN: 2504-446X. DOI: 10.3390/drones8090464
 - [53] D. Lin, J. Han, K. Li, J. Zhang, and C.-y. Zhang, "Payload transporting with two quadrotors by centralized reinforcement learning method," *IEEE Transactions on*
-

- Aerospace and Electronic Systems*, vol. 60, pp. 239–251, 2024. DOI: 10.1109/taes.2023.3321260
- [54] J. Estevez, J. Manuel Lopez-Guede, J. del Valle-Echavarri, and M. Graña, “Reinforcement learning based trajectory planning for multi-uav load transportation,” *IEEE Access*, vol. 12, pp. 144 009–144 016, 2024. DOI: 10.1109/ACCESS.2024.3470509
- [55] V. Mnih et al., “Human-level control through deep reinforcement learning,” *nature*, vol. 518, pp. 529–533, 2015.
- [56] R. S. Sutton and A. G. Barto, *Reinforcement learning - an introduction*, 2nd Edition. MIT Press, 2018. [Online]. Available: <http://www.incompleteideas.net/book/the-book-2nd.html>
- [57] F. A. Oliehoek and C. Amato, *A Concise Introduction to Decentralized POMDPs* (Springer Briefs in Intelligent Systems). Cham: Springer, 2016, ISBN: 978-3-319-28927-4. DOI: 10.1007/978-3-319-28929-8
- [58] F. A. Goodarzi and T. Lee, “Dynamics and control of quadrotor UAVs transporting a rigid body connected via flexible cables,” in *2015 American Control Conference (ACC)*, Jul. 2015, pp. 4677–4682. DOI: 10.1109/ACC.2015.7172066
- [59] E. Todorov, T. Erez, and Y. Tassa, “Mujoco: A physics engine for model-based control,” in *2012 IEEE/RSJ international conference on intelligent robots and systems*, Oct. 2012, pp. 5026–5033. DOI: 10.1109/IRoS.2012.6386109
- [60] A. Rutherford et al., “Jaxmarl: Multi-agent RL environments and algorithms in JAX,” in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons et al., Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/5aee125f052c90e326dcf6f380df94f6-Abstract-Datasets_and_Benchmarks_Track.html
- [61] R. T. Lange, *gymnax: A JAX-based reinforcement learning environment library*, version 0.0.4, 2022. [Online]. Available: <http://github.com/RobertTLange/gymnax>